

Discrete Structures

Lecture 1

Introduction

- Welcome to the Discrete Structures Course
- **Textbook(s):**
 - Discrete Mathematics and its Applications (8e) by Kenneth H Rosen
 - Contemporary Abstract Algebra (9e) by Joseph Gallian.
- Please prepare notes during lectures – Slides may not all be accessible

A Rough Grading Structure (some changes)

- Assignments: 20% [questions can appear in exams]
- Quizzes: 15%
 - Surprise quizzes are a possibility if need arises
- Mid Exam: 25%
- Final Exam: 35%
- Attendance, Participation, Surprise: 5%
- Cheating, Attendance and Ethics related issues can incur 100% penalty

Course Goals

- Introduce mathematical background needed in various branches of computer science.
- Techniques for problem solving: How to attack problems that you have never seen before.
- To convey your ideas clearly: Argue logically and write proofs formally.
- To learn abstractions and abstract reasoning.
- **ChatGPT** says Goal of this course should be to introduce the fundamental mathematical structures and proof techniques used in computer science, enabling students to model, reason about, and analyze computational problems precisely and effectively.

Overview of topics we may cover

1. Introduction to Logic & Proofs
2. Introduction to Set Theory
3. Introduction to Abstract Algebra
4. Introduction to Generating Functions
5. Introduction to Graph Theory

Proofs and Logical reasoning

- Propositions, statements
- What/why of proofs and some generic proof strategies
- Mathematical induction

Propositions

A proposition is a statement that is either **true or false** (but not both).

What is a proposition?

- $1 + 1 = 2$
- Every odd number is a prime
- It is raining (if location/time specified)
- $2^{67} - 1$ is a prime
- $(n + 1)(n - 1) = (n^2 - 1)$ for any integer n

Example of a statement that is not a proposition:

- $x + 1 = 8$

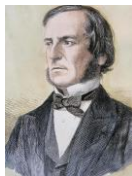
Propositional calculus



Aristotle (384 – 322 BCE)

- Propositions are statements that are either true or false.
- Just as we use variables $x, y, \dots$ for numbers, we will use variables $p, q, \dots$ for propositions.
- “if it is raining, it will be wet” : $p \rightarrow q$
- One way to combine propositions!

Propositional calculus and Boolean algebra



George Boole (1815 – 1864)

Combining propositions

- $\neg p$: It is **not** raining
- $p \vee q$: It is raining **or** there is a sprinkler overhead.
- $p \wedge q$: It is raining **and** I don't have an umbrella
- $p \rightarrow q$: If it is raining **then** it will be wet.
- If it is raining or there is a sprinkler overhead and I don't have an umbrella, then I will get wet: $((p \vee q) \wedge \neg r) \rightarrow s$.
- $p \leftrightarrow q$: p if and only if q (also written p iff q)
 - “same as” or logically equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$

Truth Tables and Logical Equivalence

p	$\neg p$
T	F
F	T

□ $\neg p$: It is not raining

Boolean OR Operator

Food Analogy: "Snacks for a party"

You're throwing a party, and you say:

"We're good if we have **pizza OR burgers**."

Pizza (A)	Burgers (B)	Party OK? (A $\vee$ B)
Yes	Yes	Yes (T)
Yes	No	Yes (T)
No	Yes	Yes (T)
No	No	No (F)

You only have a problem if **both snacks are missing**.

Boolean And Operator

💡 Analogy: "Two switches controlling one light"

Imagine a light that turns on **only if both Switch A and Switch B are ON**.

Switch A	Switch B	Light ($A \wedge B$)
ON	ON	ON (T)
ON	OFF	OFF (F)
OFF	ON	OFF (F)
OFF	OFF	OFF (F)

So, the light turns on **only if both switches are ON** — that's how Boolean AND works.

Truth Tables and Logical Equivalence

p	$\neg p$
T	F
F	T

p	q	$p \vee q$	$p \wedge q$
F	F	F	F
F	T	T	F
T	F	T	F
T	T	T	T

- $\neg p$: It is **not** raining
- $p \vee q$: It is **raining** **or** there is a sprinkler overhead.
- $p \wedge q$: It is **raining** **and** I don't have an umbrella

Boolean Implication

🍰 Analogy: "If I bake a cake (A), then I will share it with you (B)."

Let's look at each case:

1.  I bake a cake (T), I share it (T) → promise kept 
2.  I bake a cake (T), I don't share (F) → promise broken 
3.  I don't bake (F), but I share (T) → I was nice even without baking. No promise broken 
4.  I don't bake (F), and I don't share (F) → No cake, no sharing. But I made no promise, so it's still okay 

Summary:

A (If...)	B (Then...)	$A \rightarrow B$	Why
T	T	T	Promise kept
T	F	F	Promise broken
F	T	T	No promise made; no rule broken
F	F	T	No promise made; no rule broken

Truth Tables and Logical Equivalence

p	$\neg p$
T	F
F	T

p	q	$p \vee q$	$p \wedge q$	$p \rightarrow q$
F	F	F	F	T
F	T	T	F	T
T	F	T	F	F
T	T	T	T	T

- ☐ $\neg p$: It is **not** raining
- ☐ $p \vee q$: It is raining **or** there is a sprinkler overhead.
- ☐ $p \wedge q$: It is raining **and** I don't have an umbrella
- ☐ $p \rightarrow q$: **If** I am elected **then** I will lower taxes.

Boolean iff (if and only if)

■ Analogy: The Key and the Door

- **Statement:**
"The key opens the door if and only if it is the correct key."
- **Let:**
 - A = It is the correct key
 - B = The door opens

A (Correct Key?)	B (Door Opens?)	$A \leftrightarrow B$ (Key $\leftrightarrow$ Door)	Meaning
✓	✓	✓	Correct key, door opens — OK ✓
✓	✗	✗	Correct key, but door doesn't open — contradiction ✗
✗	✓	✗	Wrong key, but door opens — suspicious ✗
✗	✗	✓	Wrong key, door doesn't open — OK ✓

✓ The statement "It is the correct key if and only if the door opens" is true only when both facts are either both true or both false.

Truth Tables and Logical Equivalence

p	$\neg p$
T	F
F	T

p	q	$p \vee q$	$p \wedge q$	$p \rightarrow q$
F	F	F	F	T
F	T	T	F	T
T	F	T	F	F
T	T	T	T	T

- $\neg p$: It is **not** raining
- $p \vee q$: It is raining **or** there is a sprinkler overhead.
- $p \wedge q$: It is raining **and** I don't have an umbrella
- $p \rightarrow q$: **If** I am elected **then** I will lower taxes.

Logical Equivalence: Truth tables are identical!

- $p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$

Truth Tables and Logical Equivalence

p	$\neg p$
T	F
F	T

p	q	$p \vee q$	$p \wedge q$	$p \rightarrow q$	$p \leftrightarrow q$
F	F	F	F	T	T
F	T	T	F	T	F
T	F	T	F	F	F
T	T	T	T	T	T

- $\neg p$: It is **not** raining
- $p \vee q$: It is raining **or** there is a sprinkler overhead.
- $p \wedge q$: It is raining **and** I don't have an umbrella
- $p \rightarrow q$: If I am elected **then** I will lower taxes.

Logical Equivalence: Truth tables are identical!

- $p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$
- A number n is even **if and only if** there exists an integer k such that $n = 2k$.
- English can be imprecise, but logic is precise.

Logical operators summary

Arithmetic interpretation: T=1, F=0

Now replace **True with 1** and **False with 0**. These formulas can be evaluated using ordinary arithmetic operations.

Operator	Symbol	Arithmetic Equation	Example
NOT	$\neg p$	$1 - p$	$1 - 1 = 0$
AND	$p \wedge q$	pq	$1(1) = 1$
OR	$p \vee q$	$\min(1, p + q)$	$\min(1, 1 + 1) = 1$
XOR	$p \oplus q$	$(p + q) \bmod 2$	$(1 + 1) \bmod 2 = 0$
IMPLIES	$p \rightarrow q$	$\min(1, 1 - p + q)$	$p=1, q=0 \Rightarrow 0$
BICONDITIONAL	$p \leftrightarrow q$	$pq + (1-p)(1-q)$	$p=q=1 \Rightarrow 1$

Biconditional is exactly the negation of XOR, often called XNOR.

Predicates and quantifiers

Consider

- $\forall n \in \mathbf{N} (n + 1)(n - 1) = (n^2 - 1)$
- $\forall x, \exists y, x, y \in \mathbf{Z} x = y + 8$
- $\forall n$ stands *for all* values of n in a given domain
- $\in$ is the *element of*
- $\exists n$ stands *for exists* n
- $\mathbf{N}$ stands *for all natural numbers*
- $\mathbf{Z}$ stands *for all integers*

Some propositions are not so easy to determine e.g., $2^{67} - 1$ is not a prime.

Predicates and quantifiers

Predicates

- $x = y + 8$ is a property of x, y , so we can write, $P(x, y)$
- If we fix values to x, y it becomes a proposition.
- E.g., $P(2, 4), P(1, 7)$ are propositions.

Predicates maybe seen as propositions with holes/variables!

How can we fill the hole?

- By fixing values from the **domain**, e.g., $\mathbf{N}, \mathbf{R}$.
- By **Existential Quantification**

Used to state that **there exists at least one element** in a domain that satisfies a certain property or condition e.g., $\exists x (x^2 = 4)$.

- By **Universal Quantification**

Logical construct to assert that a **given property or condition holds for all elements** in a specific domain e.g., $\forall x (x + 5 > 10)$

Predicates and quantifiers

- Can combine and nest quantifiers:

- $\forall x(\neg(x = 0) \rightarrow \exists y(xy = 1))$

- $\forall x(L(x) \vee \exists y(L(y) \wedge F(x, y)))$ Domain = students in this class;
 $L(x)$ = x has a laptop; $F(x, y)$ = x, y are friends.

- All students in this class have a laptop or have a friend who has a laptop.

Theorems and proofs

A theorem is a proposition which can be shown true.

Contrapositive and converse:

- The **contrapositive** of “if A then B ” is “if $\neg B$ then $\neg A$ ”.
- A statement is *logically equivalent* to its contrapositive, i.e., it suffices to show one to imply the other.
- $p \rightarrow q$ is logically equivalent to $\neg q \rightarrow \neg p$
 - “If it’s raining, then the ground is wet.” ($P \rightarrow Q$)
 - “If the ground is not wet, then it is not raining.” ($\neg Q \rightarrow \neg P$)
 - Can you provide arguments in general ?
- To show “ A iff B ”, you must show two things:
 - i. A implies B **and**
 - ii. **its converse**, B implies A (OR $\neg A$ implies $\neg B$).
- Contrapositive and converse are different concepts

Contrapositive Proof

✔ Step-by-Step Proof Using a Truth Table

Let's look at the truth values of:

- $P \rightarrow Q$
- $\neg Q \rightarrow \neg P$

We'll see that **both have the same truth values in all cases**, so they are **logically equivalent**.

P	Q	$P \rightarrow Q$	$\neg Q$	$\neg P$	$\neg Q \rightarrow \neg P$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

✔ The column for $P \rightarrow Q$ and $\neg Q \rightarrow \neg P$ are **identical**.

✔ Therefore:

$$P \rightarrow Q \Leftrightarrow \neg Q \rightarrow \neg P$$

They are logically equivalent. ✔

Negating quantifiers

- What is the negation of “Every student passed the exam.”?
 - $\forall x (\text{Student}(x) \rightarrow \text{Passed}(x))$
- “There exists a student who did not pass.”
 - $\exists x (\text{Student}(x) \wedge \neg \text{Passed}(x))$

Reasoning: $\neg(\forall x P(x)) \equiv \exists x \neg P(x)$

- $\equiv$ stands for (is logically equivalent to)
- $\neg(\forall x P(x))$ says "It is not true that $P(x)$ holds for every x ."

This means:

- There is some x for which $P(x)$ is false.
- That is exactly what $\exists x \neg P(x)$ means:
- "There exists an x such that $P(x)$ is false."

Negating Quantifiers Examples

1. What is the negation of $\forall x(x^2 > x)$ $\exists x(x^2 \leq x)$
2. Show that $\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$.
3. Use predicates and quantifiers to say “Every email larger than 10MB will be compressed.” $\forall M(S(M, 10) \rightarrow C(M))$
(need to explain the notation used to be correct)

Theorems and proofs

What are the common/significant elements of the proofs?

- **Rules of inference:** Logic e.g.,
 - if p is true, and p implies q , then q is true.
 - if p is true, then $p \vee q$ is true.
 - if p is true and q is true, then $p \wedge q$ is true.
 - if p implies q and q implies r , then p implies r .
 - if $p \vee q$ is true and p is false, then q is true.
- **Strategies:** Multiple strategies/techniques such as Direct, Proof by Contrapositive or Contradiction, Vacuous, Trivial, Proof by Cases, Existence and Uniqueness Proofs, Mathematical Induction, Set Equivalence
- **Axioms:** A statement accepted as true without proof, often serving as a starting point for reasoning and building theorems e.g., Peano's axioms, Euclid's axioms

Proof Strategies/Techniques

1. Direct Proof

Structure: Assume p is true, use logic to prove q .

Form: $p \rightarrow q$

Example:

Theorem: If n is an odd integer, then n^2 is odd.

Proof: Let $n=2k+1$. Then $n^2 = 4k^2+4k+1 = 2(2k^2+2k)+1$ which is odd.

2. Proof by Contrapositive

Prove $p \rightarrow q$ by proving $\neg q \rightarrow \neg p$

Note: Sometimes easier because negations simplify expressions.

Example:

If n^2 is odd, then n is odd.

Contrapositive: If n is even, then n^2 is even

Proof Strategies/Techniques

3. Proof by Contradiction

Assume the statement you want to prove is false and show this leads to a logical contradiction.

Example: Prove $\sqrt{2}$ is irrational — classic contradiction proof.

If $\sqrt{2}$ is rational there exists p/q which is fraction in lowest terms. Square which make $p^2 = 2q^2 \Rightarrow p$ is even, so p is $2k$. $4k^2 = 2q^2 \Rightarrow q$ is even. So p/q is not in lowest terms which contradicts.

4. Vacuous Proof

When p is false, $p \rightarrow q$ is automatically true.

If the hypothesis is always false, then the implication is automatically true

Key idea: For a vacuous proof, you're always in $p = \text{false}$, so $p \rightarrow q$ is automatically true — no actual checking of q is needed.

A (If...)	B (Then...)	$A \rightarrow B$	Why
T	T	T	Promise kept
T	F	F	Promise broken
F	T	T	No promise made; no rule broken
F	F	T	No promise made; no rule broken

Proof Strategies/Techniques

5. Trivial Proof

When q is true, $p \rightarrow q$ holds regardless of p .

A (If...)	B (Then...)	$A \rightarrow B$	Why
T	T	T	Promise kept
T	F	F	Promise broken
F	T	T	No promise made; no rule broken
F	F	T	No promise made; no rule broken

6. Proof by Cases

Split the proof into separate cases (e.g., “even or odd”) and prove each individually. Exhaustive proof is a special form where each case may involve checking a finite list of possibilities e.g., Truth table.

Example: If n is an integer, then $n^2 \geq 0$.

Cases: $n > 0$, $n = 0$, $n < 0$.

Proof Strategies/Techniques

7. Existence Proofs

Constructive: Find an example explicitly.

Nonconstructive: Show existence without giving example (often contradiction or theorem).

Example: There exists a positive integer n such that $n^2-5n+6=0$.

Constructive: $n^2-5n+6=0 \Rightarrow (n-2)(n-3) = 0 \Rightarrow n=2$ or $n=3$.

Non-constructive: Per Vieta's formula sum of roots $(-b/a)$ is 5 while product is 6 (c/a) i.e., both positive and integers (additional need to argue for this), which implies a positive integer solution must exist.

8. Uniqueness Proofs

Existence: Show at least one object satisfies the property.

Uniqueness: Assume two objects satisfy the property, prove they must be the same.

Example: There is a unique real number x such that $3x+5=0$.

Existence: $x = -5/3 \Rightarrow$ at least one real x exists.

Uniqueness: $3x_1+5=0$ and $3x_2+5=0$. Subtract $3(x_1-x_2) = 0 \Rightarrow x_1 = x_2$.

Proof Strategies/Techniques

9. Proof by Mathematical Induction

Ordinary Induction: Prove for base case (say k) and an inductive step ($k+1$).

Strong Induction: Assume true for all up to k_i (k_0 till k_i), prove for $k(i+1)$.

Well-ordering principle: Every nonempty set of nonnegative integers has a least element (specifically for integers).

10. Counterexample (Disproof)

When to use: To disprove a universal claim ($\forall x, P(x)$).

Method: Find one specific example where $P(x)$ fails.

Example: "All prime numbers are odd." Counterexample: 2

11. Set Equivalence Techniques

When proving two sets are equal ($E_1=E_2$): Proving mutual inclusion ($E_1 \subseteq E_2$ and vice versa), Using set-builder notation and logical equivalences, or Employing membership tables.

Example Theorems

1. $\neg(p \wedge q)$ is logically equivalent to $\neg p \vee \neg q$
2. For all $a, b, c \in \mathbb{R}^{\geq 0}$, if $a^2 + b^2 = c^2$, then $a + b \geq c$.
3. If 6 is prime, then $6^2 = 30$.
4. x is an even integer iff $x + x^2 - x^3$ is even.
5. There are infinitely many prime numbers.
6. There exist irrational numbers x, y such that x^y is rational.
7. For all $n \in \mathbb{N}$, $n! \leq n^n$.
8. There does not exist a (input-free) program which will always determine whether an arbitrary (input-free) program will halt.

Proof for Theorem 1

Theorem 1: $\neg(p \wedge q)$ is logically equivalent to $\neg p \vee \neg q$

Proof using Truth Table i.e., Proof by cases (Exhaustive proof):

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

✓ The columns for $\neg(p \wedge q)$ and $\neg p \vee \neg q$ match exactly, so:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

Proof for Theorem 2

Theorem 2: For all $a, b, c \in \mathbb{R}^{\geq 0}$, if $a^2 + b^2 = c^2$, then $a + b \geq c$.

Direct proof as follows:

$$(a+b)^2 = a^2 + 2ab + b^2 = c^2 + 2ab \geq c^2 \text{ since } a, b \text{ are positive.}$$

Proof for Theorem 3

Theorem 3: If 6 is prime, then $6^2 = 30$.

Vacuous proof as follows:

p: 6 is prime

q: $6^2 = 30$

We are looking to prove $p \rightarrow q$

Since p is false, $p \rightarrow q$ is (vacuously) true independent of q.

Proof of Theorem 4

Theorem 4: For all $x \in \mathbb{Z}$, x is even iff $x + x^2 - x^3$ is even.

■ To show “A iff B”, you must show two things:

i. A implies B and

ii. its converse, B implies A (OR $\neg A$ implies $\neg B$).

Proof in Two directions:

■ Forward direction

■ Let $x \in \mathbb{Z}$ and x even.

■ i.e., $x = 2k$ for some $k \in \mathbb{Z}$.

■ Then $x + x^2 - x^3 = 2k + 4k^2 - 8k^3 = 2(k + 2k^2 - 4k^3)$
which is even.

■ Reverse direction

■ We will show contrapositive i.e., x is not even $\Rightarrow$

■ $x + x^2 - x^3$ is not even, i.e., x is odd $\Rightarrow x + x^2 - x^3$ is odd.

■ Let $x \in \mathbb{Z}$ be odd, i.e., $x = 2k + 1$ for some $k \in \mathbb{Z}$.

■ Then $x + x^2 - x^3$ is odd (each term is odd so sum is odd). Hence proved.

Proof for Theorem 5

Theorem 5: There are infinitely many primes.

Proof by contradiction:

1. Suppose there are only finitely many primes, say $p_1 < p_2 < \dots < p_r$.
2. Let $k = (p_1 * p_2 * \dots * p_r) + 1$.
3. When k is divided by any p_i , it has remainder 1. So none of the p_i 's divide k .
4. But $k > 1$ and k is not prime (since $k > p_r$), so k can be written as a product of primes
 - **Fundamental theorem of arithmetic:** Any natural number > 1 can be written as a (unique) product of primes.
5. Now take any prime p in this product, then, p divides k . So, by 3. above, $p \notin \{p_1, \dots, p_r\}$.
6. This contradicts 1, since we had assumed that $\{p_1, \dots, p_r\}$ was the set of all primes.



Proof for Theorem 6

Theorem 6: There exist irrational numbers x and y such that x^y is rational.

Non-constructive Proof:

- Consider $\sqrt{2}$. First show that $\sqrt{2}$ is irrational (shown earlier).
- Let $x = y = \sqrt{2}$ and consider $z = \sqrt{2}^{\sqrt{2}}$.
- Case 1: If z is rational, we are done.
- Case 2: Else z is irrational.
 - Then consider $z^{\sqrt{2}} = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = (\sqrt{2})^2 = 2$
 - Thus, we have found two irrationals $x = \sqrt{2}^{\sqrt{2}}$, $y = \sqrt{2}$ such that $x^y = 2$ is rational.

Note that the proof is not constructive since we do not determine whether $\sqrt{2}^{\sqrt{2}}$ is irrational – can we do a Constructive proof ?

Axioms

A statement accepted as true without proof, often serving as a starting point for reasoning and building theorems e.g.,

- Euclid's axioms for geometry in 300 BCE e.g., from any point to any other point, a straight line can be drawn [drawing a straight-line axiom].
- Peano's axioms for natural numbers in 1889 e.g., every natural number has a successor.
- Zermelo-Fraenkel and Choice axioms (ZFC) are a small set of axioms from which a lot of mathematics can be inferred and are widely used
 - They describe how sets behave, what kinds of sets exist etc.
 - But proving say $2+2=4$ requires Thousands of lines of proof.

Mathematical Induction

Induction

Let $P(n)$ be a property of non-negative integers. If

- $P(0)$ is true (Base case)
- For all $k \geq 0$, $P(k) \Rightarrow P(k + 1)$ (Induction Step), then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof for Theorem 7

Theorem 7: For all integers $n > 1$, $n! < n^n$

Proof by induction: we will show for all $n \geq 2$, $n! < n^n$

1. Base case For $n = 2$, $2! = 2 \leq 4 = 2^2$, so Base Case is true.
2. Induction Hypothesis: Assume, for some $n = k \geq 2$, $k! < k^k$
3. Induction step: To show: $(k + 1)! \leq (k + 1)^{(k+1)}$
$$(k + 1)! = k! \cdot (k + 1) \leq k^k (k + 1) \text{ (by Induction Hypothesis)}$$
$$< (k + 1)^k \cdot (k + 1) = (k + 1)^{(k+1)}$$
4. Hence by induction, we conclude that for all $n \geq 2$, $n! < n^n$.

Examples by induction

- Summation: For every positive integer n ,

$$1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1}n^2 = (-1)^{n-1} \frac{n(n+1)}{2}$$

- Inequality: If $h > -1$, then $1 + nh \leq (1 + h)^n$ for all non-negative integers n .

Strong Induction

Theorem: Every integer $n > 1$ can be written as a product of prime numbers.

- If you try **weak induction**, you only get to assume the statement for $n=k$ when proving it for $n=k+1$.
- If $n+1$ is composite, you'll need the property for **both** of its smaller factors — which could be much smaller than n .
- Hence need strong induction: you assume the property for *all* numbers less than your target, so you can apply it to *any* factorization step.

Strong Induction

Proof for “Every integer $n > 1$ can be written as a product of prime numbers.”

- **Base case:** $n=2$ is prime, so it's a product of primes (just itself).
- **Inductive hypothesis:** Assume that every integer m with $2 \leq m \leq k$ can be written as a product of primes.
- **Inductive step:** Consider $k+1$.
- If $k+1$ is prime, we are done.
- If $k+1$ is composite, then $k+1 = a \cdot b$ where $2 \leq a, b \leq k$.
- By the inductive hypothesis, both a and b can be written as products of primes.
- Multiplying these prime products gives a prime factorization for $k+1$.
- Thus, by strong induction, every integer $n > 1$ is a product of primes.

Strong Induction vs. WOP

Strong induction:

- You fix a number n and assume the statement holds for all smaller positive integers.
- Then you prove it holds for n itself.
- This moves upward from the base case step-by-step.

Well-Ordering Principle:

- You argue downward: assume there is a counterexample, take the smallest counterexample, and derive a contradiction.
- You do not explicitly “step through” all numbers in order — you jump to the “first bad case” and show it can’t exist.

Well Ordering Principle (WOP)

Theorem: Every integer $n > 1$ can be factored into primes.

Proof (via WOP):

Suppose not. Let S be the set of integers > 1 that are not products of primes.

If S is nonempty, WOP says it has a smallest element m [since WOP states “Every nonempty set of natural numbers has a least element.”]

- m cannot be prime – if prime it is a product of $1 \cdot \text{prime}$ (itself).
- So, m is composite: $m = a \cdot b$ with $1 < a < m$ and $1 < b < m$.
- Since m is the **smallest** element of S , neither a nor b can belong to S .
- Therefore both a and b can be factored into primes. Multiplying them gives a prime factorization for m — Contradiction.

Thus, S must be empty, so every integer $n > 1$ is a product of primes.

Proof for Theorem 8

Theorem 8: There does not exist a (input-free) program which will always determine whether an arbitrary (input-free) program will halt [Halting Problem].

More formally: There is no algorithm H that, given the description of *any* program P with no inputs, always correctly answers “halts” or “does not halt” and itself always halts.

Proof by Contradiction:

Suppose such a program $H(P)$ exists:

- Input: code of a program P (no input to P itself)
- Output: YES if P halts, NO if P runs forever
- H itself always halts.

Proof for Theorem 8

- Build a function as follows:

function P():

 if H(P) == YES:

 loop forever

 else:

 halt immediately

- P runs H on *its own code*.
- If H predicts P halts $\rightarrow$ P runs forever.
- If H predicts P runs forever $\rightarrow$ P halts.

Proof for Theorem 8

Contradiction

Now ask: what does $H(P)$ output?

- If $H(P) = \text{YES}$ (predicts halt) $\rightarrow$ by definition, P loops forever $\rightarrow$ prediction wrong.
- If $H(P) = \text{NO}$ (predicts infinite loop) $\rightarrow$ P halts immediately $\rightarrow$ prediction wrong.

Either way, H makes a wrong prediction for P , contradicting our assumption that H works for all programs.

Hence, no such H can exist. Proves:

"Halting Problem" is undecidable even for input-free programs.

